

Cybersecurity Basics - Learning notes (Day 1)

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1. SIX IMPORTANT THINGS I LEARNED

- I learned that cybersecurity is the practice of securing data, software, hardware, and devices connected to the internet.

- I learned about threat, vulnerability and risk.

Threat: is any potential danger that could exploit a system's weakness or vulnerability.

Vulnerability: is any weakness or flaws in a system that could be exploited by a threat.

Risk: is the likelihood that a threat will exploit a vulnerability and the impact it could have.

- I learned what the CIA Triad is all about. The acronym CIA means Confidentiality, Integrity, and Availability. They are the Fundamentals of cybersecurity.

Confidentiality: This is the practice of ensuring that sensitive data or information are only accessed by authorized individuals or systems. Confidentiality is achieved with the use of encryption, access control lists (ACLs), multifactor authenticator (MFA), data classification etc.

Integrity: This is the practice of that data or information remaining in its original state, without being altered or modified intentionally or accidentally. Data integrity is achieved with the use of Hashing, Read-Only User access permissions, digital signatures etc.

Availability: This is the practice of ensuring that data or network resources are available to authorized individuals or systems when they need it without delay or interruption. This is achieved through the use of redundancy, data backups, disaster recovery plans, regular system maintenance etc.

- I learned about encryption and the types. Encryption is the process of converting data (plaintext) into unreadable format (ciphertext) using algorithm and a key, so only the authorized user can decrypt/convert the ciphertext into a readable format. There are two main types of encryption, which are symmetric and asymmetric.

Symmetric: This type of encryption uses a single key to encrypt and decrypt data. It is fast and efficient for large amount of data. Examples are DES, 3DES, AES etc

Asymmetric: This type of encryption uses two keys, one for encryption and the other for decryption. They are called the public and private keys. It is more secure but slower than symmetric, it's not suitable for large amount of data. It is mostly used for digital signatures. Examples are: RSA and DSA

- I learned about the TCP/IP model. The TCP/IP model is a practical model that explains how data moves across networks. It is a simplified model of the OSI models. There are 4 layers in the TCP/IP Model namely;

Application Layer: This layer provides services and protocol that allow the user application to access network resources and communicate over the network. Examples of protocols in this layer are: DNS,HTTP,FTP,SMTP etc

Transport Layer: This layer is responsible for end-to-end communication between applications running on different hosts. It ensures reliable data transfer, segmentation, flow control and error checking. Examples of protocols in this layer is ; TCP and UDP.

Internet Layer: This layer is responsible for logical addressing(IP address), and routing of packets across different networks. The protocols in this layers are ; ICMP, IP, ARP(Ipv4), NDP(IPv6).

Network Access Layer: This layer handles the physical connection of devices to a network and how data are transmitted within local networks including framing and mac addressing.

- I learned what a SOC analyst is, and their roles. A Security Operation Center (SOC) is a team of cybersecurity professionals that protects and monitors an organization's networks for security events.

A SOC analyst is responsible for monitoring and analyzing security events, detecting and investigating security threats, and responding to or recovering from security incidents.

2. FIVE QUESTIONS I STILL HAVE

- What type of cyber attacks and alerts do SOC Analysts deal with most frequently?
- How can I clearly differentiate between a threat, a vulnerability, and a risk during real-world incidents?

- Which tools are used daily by SOC Analysts, and how do they help in identifying cyber attacks?
- How do SOC Analyst detect and triage threats in real-time inside a SOC environment?
- How does encryption actually work during communication between two systems?

3. WHY I WANT TO BECOME A SOC ANALYST

I want to become a SOC Analyst because I am naturally curious, enjoy solving problems and want to play an important role in defending organizations from cyber threats. I like the challenges of investigating suspicious activity and understanding how attackers think. Becoming a SOC analyst will allow me to use my technical skills to protect businesses, people, and critical systems. I want a career where I can keep learning, stay updated, and contribute to making digital environments safer.